

In-Processing vs Post-Processing Proof

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Definitions

Definition 1. Let (X, S, Y) be random variables with sensitive attribute $S \in \{0, 1\}$. Let model f produces a score $T = f(X) \in [0, 1]$. For each $s \in \{0, 1\}$, we can define the conditional CDF as

$$F_s^f(t) := \Pr(T \leq t \mid S = s), \quad t \in [0, 1].$$

Let $F^f := (F_0^f, F_1^f)$.

Definition 2. Let the area between CDF curves of a model f be

$$\text{ABCC}(f) := \int_0^1 |F_0^f(t) - F_1^f(t)| dt.$$

$\implies \text{ABCC}(f) = 0$ if and only if $F_0^f = F_1^f$ almost everywhere.

Definition 3. We can fix the thresholds $0 \leq t_1 < \dots < t_K \leq 1$ and bounded linear functionals $M_1, \dots, M_L$. Let the finite DP summary of f be

$$\phi(F^f) := (F_0^f(t_1) - F_1^f(t_1), \dots, F_0^f(t_K) - F_1^f(t_K), M_1(F^f), \dots, M_L(F^f)) \in \mathbb{R}^{K+L}.$$

Definition 4. An in-processing method chooses f by minimizing

$$J(f) := \mathcal{L}_{\text{sup}}(f) + \lambda R(\phi(F^f)),$$

where $\mathcal{L}_{\text{sup}}$ is a supervised loss, $\lambda \geq 0$, and $R : \mathbb{R}^{K+L} \rightarrow \mathbb{R}$ is arbitrary. The fairness term depends on f only through $\phi(F^f)$.

Main Theorem

Theorem 1. Let J be any in-processing objective of the above form. Then there exist a data distribution for (X, S, Y) and two models f_1, f_2 such that:

1. $\mathcal{L}_{\text{sup}}(f_1) = \mathcal{L}_{\text{sup}}(f_2)$,
2. $\phi(F^{f_1}) = \phi(F^{f_2})$ (hence $J(f_1) = J(f_2)$),
3. $\text{ABCC}(f_1) \neq \text{ABCC}(f_2)$.

This means minimizing J does not minimize ABCC, so in-processing DP methods are not ABCC-consistent.

Proof

Let $\mathcal{D}$ be the set of all pairs of CDFs $F = (F_0, F_1)$ on $[0, 1]$. Let

$$\text{ABCC}(F) := \int_0^1 |F_0(t) - F_1(t)| dt.$$

Lemma 1. *The map $\phi : \mathcal{D} \rightarrow \mathbb{R}^{K+L}$ is not injective. Also, $\exists F, G \in \mathcal{D}$ such that*

$$\phi(F) = \phi(G) \quad \text{but} \quad \text{ABCC}(F) \neq \text{ABCC}(G).$$

Proof. We can take any $F = (F_0, F_1) \in \mathcal{D}$ with $F_0 \not\equiv F_1$. Then we can choose two disjoint open intervals $I_1, I_2 \subset (0, 1)$ that contain none of the thresholds $t_1, \dots, t_K$, and also where $F_0 \neq F_1$ intersects both intervals in sets of positive measure.

We can then pick smooth functions ψ_1, ψ_2 with $\text{supp}(\psi_j) \subset I_j$, $\int \psi_j \neq 0$, and choose so that

$$M_\ell(0, \psi_1) = M_\ell(0, \psi_2) \quad \text{for each } \ell = 1, \dots, L.$$

Let, for small $\alpha \in \mathbb{R}$,

$$G_1^{(\alpha)}(t) := F_1(t) + \alpha\psi_1(t) - \alpha\psi_2(t), \quad G^{(\alpha)} := (F_0, G_1^{(\alpha)}).$$

For sufficiently small α , $G_1^{(\alpha)}$ remains a valid CDF, and because $\psi_1(t_k) = \psi_2(t_k) = 0$,

$$F_0(t_k) - F_1(t_k) = F_0(t_k) - G_1^{(\alpha)}(t_k) \quad \text{for all } k.$$

Similarly,

$$M_\ell(G^{(\alpha)}) = M_\ell(F) \quad \text{for all } \ell,$$

so $\phi(G^{(\alpha)}) = \phi(F)$ for all small α .

But,

$$\text{ABCC}(G^{(\alpha)}) = \int_0^1 |F_0(t) - G_1^{(\alpha)}(t)| dt$$

changes with α because the integrand is distributed on sets of positive measure. $\implies \text{ABCC}(G^{(\alpha)})$ is a nonconstant continuous function of α , so for $\alpha_1 \neq \alpha_2$ we get that

$$\text{ABCC}(G^{(\alpha_1)}) \neq \text{ABCC}(G^{(\alpha_2)}), \quad \phi(G^{(\alpha_1)}) = \phi(G^{(\alpha_2)}) = \phi(F).$$

Take $F := G^{(\alpha_1)}$, $G := G^{(\alpha_2)}$. □

Let $F, G \in \mathcal{D}$ satisfy $\phi(F) = \phi(G)$ but $\text{ABCC}(F) \neq \text{ABCC}(G)$. We can then assume the hypothesis class is rich enough to realize these conditional CDFs: $\exists f_1, f_2$ such that

$$F^{f_1} = F, \quad F^{f_2} = G.$$

Then, we can construct a data distribution for (X, S, Y) where the joint distribution of (Y, T) is the same under f_1 and f_2 . We can do this by permuting the

scores of f_1 within each label class $Y = y$ so that the conditional distributions given S become G . Since the permutation is label-preserving,

$$\mathcal{L}_{\text{sup}}(f_1) = \mathbb{E}[\ell(Y, f_1(X))] = \mathbb{E}[\ell(Y, f_2(X))] = \mathcal{L}_{\text{sup}}(f_2)$$

for any pointwise supervised loss ℓ .

Because $F^{f_1} = F$ and $F^{f_2} = G$,

$$\phi(F^{f_1}) = \phi(F) = \phi(G) = \phi(F^{f_2}),$$

$\implies$

$$J(f_1) = \mathcal{L}_{\text{sup}}(f_1) + \lambda R(\phi(F^{f_1})) = \mathcal{L}_{\text{sup}}(f_2) + \lambda R(\phi(F^{f_2})) = J(f_2).$$

But,

$$\text{ABCC}(f_1) = \text{ABCC}(F) \neq \text{ABCC}(f_2) = \text{ABCC}(G).$$

This means, two models have identical in-processing objective value but different ABCC. This means, optimizing J does not imply optimizing ABCC. $\square$